

## Chapter 3.7 – Learning Frameworks

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PAC Learning • Estimating Hypothesis Accuracy • Hoeffding's Inequality • VC Dimension

# Learning Objectives

Explain the role of computational learning theory in analyzing machine-learning algorithms.

Describe PAC (Probably Approximately Correct) learning.

Understand sample complexity and the finite/infinite hypothesis-space distinction.

Estimate hypothesis accuracy using sample error and confidence intervals.

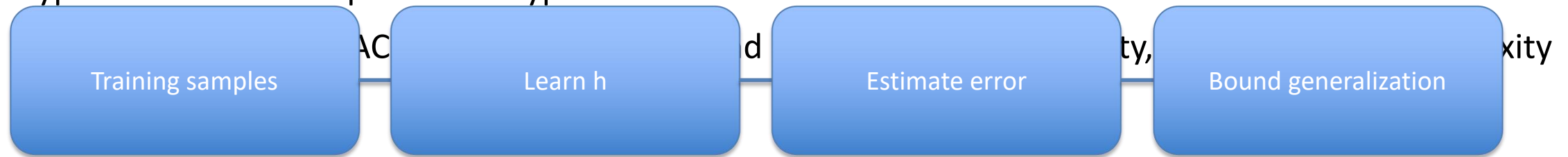
Explain the relationship between in-sample and out-of-sample error using Hoeffding's inequality.

Define VC dimension and explain the idea of shattering.

# Why a Learning Framework?

A learner may fit the available training examples well but still misclassify unseen instances.

A theoretical framework helps answer: How many examples are enough? How accurate is the learned hypothesis? How complex is the hypothesis class?



# PAC Learning – Core Idea

PAC = Probably Approximately Correct.

A hypothesis that makes wrong predictions most of the time can be rejected.

A hypothesis that is consistently correct on a large number of samples is unlikely to be wrong.

“Approximately” refers to being close to the target function; “probably” refers to having low generalization error with high confidence.

An algorithm that returns a PAC hypothesis is called a PAC algorithm.

# PAC Framework: Basic Assumptions

1. Unseen samples are drawn according to a fixed distribution related to the learning problem.
2. The proportion of wrongly classified samples is the error rate.
3. The learner seeks a hypothesis  $h$  that is close to an unknown target function  $f$ .
4. A good hypothesis should generalize beyond the observed training samples.

# PAC Error Parameters $\epsilon$ and $\delta$

$\epsilon$  (epsilon) represents the tolerated approximation/generalization error.

$\delta$  (delta) represents the small probability that the learned hypothesis fails to meet the desired error bound.

PAC objective: obtain a hypothesis with error  $\leq \epsilon$  with probability at least  $1 - \delta$ .

**Goal:  $P(\text{error}(h) \leq \epsilon) \geq 1 - \delta$**

Smaller  $\epsilon$  demands higher accuracy; smaller  $\delta$  demands greater confidence.

*$\epsilon$  = accuracy tolerance;  $\delta$  = allowed failure probability*

# PAC Sample Complexity

For a finite hypothesis space, the book derives a sample-size condition from the probability that an incorrect hypothesis remains consistent with the observed samples.

The probability that a particular bad hypothesis agrees with  $N$  samples decreases exponentially with  $N$ .

Considering all hypotheses leads to a bound involving  $|H|$ ,  $\epsilon$  and  $\delta$ .

**$N$  grows with  $\ln |H|$ ,  $\ln(1/\delta)$ , and  $1/\epsilon$**

The book expresses the resulting sample-complexity relationship in logarithmic form.

*More hypotheses or stricter accuracy/confidence requirements require more training samples.*

# Worked Interpretation of PAC Sample Complexity

If the hypothesis space becomes larger, more samples are required to identify a reliable hypothesis.

If  $\epsilon$  is reduced, the learner demands a smaller generalization error, so the required sample size increases.

If  $\delta$  is reduced, the learner demands greater confidence, so the required sample size increases.

Thus, sample complexity captures the data requirement for reliable generalization.

# Mistake Bound Model

The mistake-bound model is an alternative learning framework to PAC.

The learner is evaluated by the number of prediction mistakes made before converging to the correct hypothesis.

Unlike PAC, the learner predicts the target function before being shown the correct label and the mistakes are counted.

This is useful for applications where mistakes are costly and the number of mistakes before learning matters.

# Mistake Bounds: Key Ideas

For Find-S, the book discusses a bound related to the number of hypothesis variables.

For Find-S and a related learning method, the text gives logarithmic-style mistake bounds.

For an arbitrary concept class  $C$ , the worst-case number of mistakes is related to its VC dimension and logarithmic terms.

Important distinction: PAC focuses on probability/error guarantees; mistake-bound learning focuses on the count of prediction errors before convergence.

# Finite vs Infinite Hypothesis Spaces

## Finite Hypothesis Space

PAC framework can identify classes learnable from a polynomial number of training samples.

Sample complexity can be related to  $|H|$ .

Number of possible hypotheses is finite.

## Infinite Hypothesis Space

A direct finite- $|H|$  bound is not sufficient.

VC dimension provides a way to characterize the complexity of the hypothesis class.

Generalization bounds depend on the ability of the class to shatter finite sets.

# Sample Complexity vs Computational Complexity

## Sample Complexity

Sample complexity: number of training instances required for a learner to converge to a successful hypothesis.  
How much data?

Generalization/data efficiency

A method can have a manageable sample requirement but still be computationally expensive.

## Computational Complexity

How much computation?

Computational complexity: amount of computational effort required for the learner to converge.  
Time/effort required for

Algorithmic efficiency

Both dimensions matter when assessing the practicality of a learning framework.

## 3.7.2 Estimating Hypothesis Accuracy

Sample error of  $h$ : proportion of training instances misclassified by  $h$ .

True error of  $h$ : probability that  $h$  misclassifies a randomly drawn instance from the domain.

Training performance can be better than true performance because the model has already seen the training examples.

$$\text{Sample Error} = (\# \text{ misclassified training instances}) / n$$

Confidence intervals can be used to estimate the true error from the observed sample error.

$$\text{True Error} = P[h(x) \neq f(x)] \text{ for a random domain instance}$$

# Confidence Interval for Hypothesis Error

The book presents a 95% confidence-style approximation using the sample error and sample size.

The expression has the form sample error  $\pm 1.96$  times a standard-error term.

Larger  $n$  reduces the uncertainty around the estimate.

This is an estimation tool: it does not make the sample error itself equal to the true error.

$$\text{error}(h) \pm 1.96 \sqrt{[\text{error}(h)(1-\text{error}(h))]/n}$$

*95% confidence-style interval presented in the text*

# In-Sample vs Out-of-Sample Error

## In-Sample

In-sample error / empirical error: difference between target function  $f$  and hypothesis  $h$  on training data  
Observed on training data

## Out-of-Sample

Observed on unseen data

Out-of-sample error / generalization error: difference between  $f$  and  $h$  on unknown test data.

Usually easier to measure

Generalization error

Key indicator of real predictive ability

A good learning framework seeks a small gap between training and generalization performance.

The book introduces Hoeffding's inequality to quantify a probabilistic bound on this deviation.

### 3.7.3 Hoeffding's Inequality

Hoeffding's inequality provides a probabilistic upper bound on the deviation between in-sample and out-of-sample errors.

It is presented for a finite hypothesis set.

The deviation is bounded by a term that decreases exponentially as the sample size increases.

$$P(|E_{\text{in}} - E_{\text{out}}| > \epsilon) \leq 2e^{-2\epsilon^2 n}$$

Therefore, larger training samples make the observed error a more reliable indicator of generalization error.

*$\epsilon > 0$ ;  $n$  = sample size; the bound decreases rapidly as  $n$  increases.*

# Interpreting Hoeffding's Bound

If  $n$  increases, the probability of a large difference between empirical and true error decreases.

If  $\epsilon$  is made smaller, the desired error gap is tighter, making the guarantee more demanding.

The inequality is a theoretical guarantee rather than a new training algorithm.

It provides a mathematical reason for why more representative training data improves confidence in generalization.

### 3.7.4 Vapnik–Chervonenkis (VC) Dimension

VC dimension is used to characterize the complexity of a hypothesis/classifier, especially for an infinite hypothesis set.

VC dimension is the maximum number of data points that can be shattered by the hypothesis class.

Shattering means correctly realizing all possible binary labelings of those points.

**$VC(H) = \text{maximum } n \text{ such that } H \text{ can shatter } n \text{ points}$**

For  $n$  points, there are  $2^n$  possible positive/negative labelings.

*Shattering  $n$  points means all  $2^n$  binary labelings can be realized.*

# Shattering – 2 Points

For two points, a linear classifier can realize all  $2^2 = 4$  binary labelings under suitable point placement.

The concept is not simply separating one particular labeling; the classifier class must realize every possible labeling of the chosen points.

This illustrates the basic idea behind VC dimension.

# Shattering – 3 Points

For three suitable points, a linear classifier can realize all  $2^3 = 8$  binary labelings in the example discussed by the book.

The ability to shatter more points indicates a more expressive hypothesis class.

Other representations such as planes, hyperplanes, squares, rectangles and circles can shatter larger sets under suitable conditions.

# VC Dimension and Model Complexity

More expressive/complex hypothesis classes generally have higher VC dimension.

The book mentions perceptrons and SVMs as examples of complex learners capable of classifying more data configurations.

VC dimension is used to derive generalization bounds for infinite hypothesis spaces.

Model complexity therefore affects how much training data is needed for reliable generalization.

# Learning Frameworks – Integrated View

PAC learning asks whether a learner can find a hypothesis that is probably approximately correct.

Sample complexity asks how many examples are needed.

Computational complexity asks how much computation is needed



## Exam-oriented Summary – Section 3.7

Define PAC learning and explain  $\epsilon$  and  $\delta$ .

Derive/interpret sample-complexity dependence on  $|H|$ ,  $\epsilon$  and  $\delta$ .

Differentiate PAC and mistake-bound models.

Differentiate finite and infinite hypothesis spaces.

Define sample error and true error.

Explain Hoeffding's inequality and the role of  $n$ .

Define VC dimension and explain shattering using 2-point/3-point examples.